

Sandeep Regmi

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Education

Ph.D. in Transportation Engineering University of Arizona, Tucson, AZ Advisor: Yao-Jan Wu, Ph.D., P.E.	Aug. 2025 – Present
Bachelor of Civil Engineering Purwanchal Campus, Institute of Engineering, Tribhuvan University, Nepal	Nov. 2018 – Apr. 2023

Research Interests

- **Intelligent Transportation Systems (ITS)**
 - Traffic detection and incident management
 - Traffic operations and simulation
 - Traffic management systems and software lifecycle
- **Data-Driven Transportation Systems Analysis**
 - Transportation data management and analysis
 - Traffic sensor integration and dashboard applications
 - Transportation system performance measurement
- **Transportation Safety Analysis and Crash Modeling**
 - Automated vehicle safety assessment

Work Experience

Graduate Research Assistant Center for Applied Transportation Sciences (CATS) University of Arizona	Dec. 2025 – Present
Graduate Research Assistant Arizona Transportation Institute (AZTI) University of Arizona	Aug. 2025 – Dec. 2025
Civil Engineer China State Construction Engineering Corporation Limited Nawalparasi (West), Nepal	July 2023 – May 2025

Publications

1. Regmi, S., Pudasaini, P., & Wu, Y.-J. (2026). Early Detection and Localization of Unreported Freeway Incidents Using Multisource Traffic Data. Accepted for presentation at the 106th TRB Annual Meeting 2027, Washington, DC, Jan. 10–14, 2027. TRBAM-27-05694.

Project Experience

Safety Assessment of Automated Vehicles in the City of Phoenix	Aug. 2026 – Present
• Assist with acquiring and integrating automated-vehicle crash, operational, roadway, and traffic datasets into a unified geospatial and analytical database. • Support comparative analyses of automated and human-driven vehicle crash rates, severity, typologies, and contributing roadway and environmental factors.	

NCHRP 03-152: Managing the Life Cycle of Software for Evolving Traffic Management Systems June 2026 – Present

- Support a state-of-the-practice review of software lifecycle management for traffic management systems, covering planning, procurement, integration, testing, maintenance, and staffing.
- Assist with developing a nationwide agency survey to document current practices, challenges, and emerging strategies across state DOTs and regional and local agencies.

HAWK Pedestrian Signal Warrant Review (CATS TSR FY26-05, PCDOT) Summer 2026

- Reviewed HAWK and pedestrian-signal warrant criteria used by agencies in Arizona and other states, including potential updates in the 11th Edition of the MUTCD.
- Summarized agency practices in a comparison table and developed recommendations for updating PCDOT's approach, with attention to latent pedestrian demand and volume thresholds.

Development of Integrated Traffic Data Dashboard for Peoria Feb. 2026 – Present

- Clean and analyze data from heterogeneous traffic sensors to support data integration, database development, and dashboard modules.
- Assist with reviewing detector inventories, vendor data structures, and API capabilities for modules addressing volume, speed, arrivals on green, and related performance measures.

PTZ Cameras Literature Review (CATS TSR FY26-01, TDTM) Fall 2025

- Surveyed Arizona transportation agencies and vendors to catalog approved pan-tilt-zoom camera products, specifications, and traffic-related capabilities.
- Collected and synthesized camera-sharing policies and national best practices for coordination with law enforcement and partner agencies, informing TDTM's consideration of PTZ cameras as standard ITS devices.

Professional Presentations

Conference Presentations

1. Early and Unreported Roadway Incident Detection Using Multisource Traffic Data. Poster presented at the 2026 ASCE International Conference on Transportation & Development, June 28–July 1, 2026.
2. Early and Unreported Roadway Incident Detection Using Multisource Traffic Data. Poster presented at the 2026 Mountain District Annual Meeting, June 10–12, 2026.

Awards and Honors

- **Bachelor of Civil Engineering Partial Scholarship:** Institute of Engineering entrance examination (ranked 1,299 among approximately 16,000 examinees), 2017.
- **Higher-Secondary Partial Scholarship:** School Leaving Certificate and entrance-examination performance, 2016.
- **Annual Full Scholarship:** awarded for outstanding academic performance throughout school.

Professional Membership & Leadership

Publicist , University of Arizona ITE Student Chapter	Aug. 2026 – Present
Member , Institute of Transportation Engineers (ITE) Arizona Section	May 2026 – Present
Student Member , American Society of Civil Engineers (ASCE)	Feb. 2026 – Present
Member , Intelligent Transportation Society of Arizona (ITS Arizona)	Oct. 2025 – Present

Licenses

- Registered Civil Engineer, Nepal Engineering Council, Civil Category “A,” Registration No. 80231.